

Canonical Methods for Avoiding Catastrophic Forgetting in Deep RL

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Contents

Canonical Methods for Avoiding Catastrophic Forgetting in Deep RL	2
Executive summary	2
Early documentation	2
What should be measured	2
Canonical method families	3
Experience replay	3
Parameter regularization and synaptic consolidation	4
Generative replay and pseudo-rehearsal	4
Architectural isolation and modularity	5
Distillation and progress followed by compression	5
Gradient constraints	5
Practical method selection	5
Open limitations	6
References	7

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Executive summary

Deep reinforcement learning agents can lose previously learned behavior when their parameters are updated on new tasks, new environments, or later regions of an otherwise unchanged environment. The standard term **catastrophic forgetting** usually refers to a drop in performance on an earlier task after learning a later one. Deep RL also exhibits two related problems that require different remedies:

- **Within-task catastrophic interference:** learning later states or subgoals damages predictions needed for earlier parts of the same MDP [Fed20].
- **Task conflict:** two tasks require different actions from the same observation, so a single task-agnostic policy cannot represent both solutions [Kes21].

The canonical defenses fall into four broad families: replay, parameter consolidation, generative replay, and architectural isolation. In practice, replay is the strongest first baseline; regularization is attractive when memory is limited; modular or multi-head architectures are needed when objectives conflict; and progressive architectures provide the strongest retention guarantees at the cost of model growth.

Early documentation

The broad neural-network literature had documented catastrophic forgetting before the deep-RL era. An early reference is the study of rehearsal mechanisms in neural networks [Rob93].

For modern neural function approximation in reinforcement learning, one of the earliest direct treatments surfaced in this workspace is Goodrich’s work on catastrophic forgetting in temporal-difference learning [Goo15b]. It identifies correlated, nonstationary online data and overlapping hidden representations as sources of interference, and proposes localizing updates through selective neuron activation. The exact historical priority of this line of work should be treated cautiously because the full text is not currently available.

The most influential early deep-RL reference is the EWC paper [Kir16], first released on **December 2, 2016** and later published in PNAS in **2017**. It demonstrated sequential learning across Atari games and introduced a practical fixed-capacity method for protecting parameters important to old tasks.

What should be measured

A report of forgetting should not rely only on final reward. At minimum, it should report:

- **Retention:** performance on each previous task before and after later training.
- **Backward transfer:** whether learning a new task improves or harms old tasks.
- **Forward transfer:** whether earlier tasks make new tasks faster to learn.
- **Plasticity:** speed and final performance on the current task.
- **Memory and compute:** replay storage, auxiliary parameters, and training overhead.

- **Task information:** whether task boundaries, task identities, or context labels are available.

These distinctions are central to continual-RL evaluation [Khe20].

Canonical method families

Family	Core idea	Representative references	Best fit	Main weakness
Experience replay	Reuse old transitions while learning new data	[Rol18, Ise18]	Task-agnostic sequences with compatible objectives	Memory and off-policy costs
Parameter consolidation	Penalize changes to important parameters	[Kir16, Zen17, Alj17]	Fixed-size models and identifiable tasks	Stability can reduce plasticity
Multi-timescale consolidation	Store parameter history at fast and slow timescales	[Kap18, Kap19]	Continuously changing environments	Extra computation and weaker large-scale evidence
Generative replay	Recreate old states or trajectories with a generator or world model	[Atk18, Ket19b]	Limited raw-data storage	Generator drift and compounding errors
Modularity and isolation	Freeze old modules or use task-specific heads	[Rus16, Kes21]	Conflicting objectives or strong retention requirements	Capacity growth and task selection
Distillation	Preserve old policies or values as soft targets	[Sch18b, Tra19]	Consolidating multiple policies	Can preserve outdated or incompatible behavior
Gradient constraints	Restrict new updates using gradients on old memories	[Lop17b]	Small episodic memories and explicit retention constraints	Computationally expensive in large networks

Experience replay

Replay is the simplest and most broadly useful defense. CLEAR combines on-policy experience with replayed historical experience, using off-policy correction and behavioral cloning losses to stabilize both policy and value predictions [Rol18]. It does not require explicit task identities or task boundaries and substantially reduces forgetting in Atari and DMLab.

Selective replay keeps a long-term memory in addition to a standard recent-experience buffer. Distribution matching was more reliable than selection based only on reward, surprise, or state-space coverage in

the studied lifelong-learning settings [Ise18]. This suggests that replay quality matters: a buffer should preserve coverage of old behavior, not merely unusual or high-reward transitions.

Replay is usually the correct first baseline because it directly rehearses the function that must be retained. Its limits are equally important. Large buffers consume memory, replay changes the data distribution for on-policy methods, and a shared buffer cannot resolve contradictory labels or reward functions. In the latter setting, replay can repeatedly expose the model to incompatible targets rather than solve the conflict [Kes21].

Parameter regularization and synaptic consolidation

EWC protects old knowledge by estimating which parameters matter for a previous task and penalizing movement away from their old values:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{new}}(\theta) + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2.$$

Here, F_i is typically the diagonal Fisher information, θ_i^* is the old parameter value, and λ controls the stability–plasticity tradeoff [Kir16]. In sequential Atari experiments, EWC allowed a single network to retain knowledge across multiple games, although it remained below the performance of separate networks and required extra importance statistics.

Online importance methods such as Synaptic Intelligence and Memory Aware Synapses estimate parameter importance during or after learning rather than relying only on a task-end Fisher estimate [Zen17, Alj17]. They are useful conceptual baselines, but most evidence for these variants comes from general continual-learning benchmarks rather than large-scale RL.

Complex synapses and policy consolidation extend the idea to continuous, multi-timescale memory. Complex synapses place each visible parameter at the front of a chain of slower variables, allowing recent learning to remain plastic while older information is consolidated gradually [Kap18]. Policy consolidation uses a cascade of hidden policies that regularize the current policy by its history and does not require known change timescales [Kap19]. These methods are attractive for task-agnostic nonstationarity, but the complex-synapse experiments also report computational overhead and difficulty matching control agents on larger Atari-style tasks [Kap18].

Generative replay and pseudo-rehearsal

Generative replay replaces stored old data with samples produced by a learned generator. The general deep-generative-replay framework trains a generator and a task-solving network together, then interleaves generated old-task samples with data from the current task [Shi17].

RePR adapts this idea directly to deep RL. A short-term DQN learns the current Atari game, while a long-term DQN is consolidated through distillation and pseudo-rehearsal from a generative model [Atk18]. The method avoids storing raw data from every previous game and retained strong performance on the reported Atari sequences.

The tradeoff is that the generator becomes a second continual-learning problem. As the task sequence grows, generated samples may lose fidelity or underrepresent older tasks. World-model pseudo-rehearsal uses internally generated rollouts for a related purpose and can reduce forgetting in learned representa-

tions without explicit task labels [Ket19b]. Generative replay is therefore most compelling when raw-data storage is costly and a stable world or state model is already available.

Architectural isolation and modularity

Progressive Neural Networks prevent forgetting by freezing old task-specific columns and adding a new column for each task. Lateral connections allow the new column to reuse earlier features without modifying the computations that produced old behavior [Rus16]. This gives a strong retention guarantee by construction and produced positive transfer across Atari and 3D-maze tasks.

The cost is capacity growth. Progressive networks add parameters with each task, and the task identity or active column must be known or inferred. Related modular methods use separate heads, masks, or subnetworks to isolate conflicting behavior while sharing lower-level representations.

OWL makes the task-conflict issue explicit. It uses shared features with separate task-specific policy heads, applies EWC to the shared representation, and selects among heads with a bandit-like procedure when the task is unknown [Kes21]. This combination is important because separate heads solve objective conflict, while regularization protects shared features from representational forgetting.

Distillation and progress followed by compression

Policy distillation transfers several previously learned policies into a consolidated policy or representation. DisCoRL combines state-representation learning with policy distillation for continual navigation and can infer which behavior is appropriate without an external task signal [Tra19].

Progress and Compress separates rapid learning of the current task from slower consolidation into a knowledge base. The active component learns the new task, after which its knowledge is compressed into a persistent component while protecting earlier skills [Sch18b]. These methods are useful when a learner needs both fast adaptation and a stable long-term memory, but the consolidation objective must be designed carefully when policies are incompatible.

Gradient constraints

Gradient Episodic Memory stores a small memory of old examples and projects the current gradient so that the update does not increase loss on those memories [Lop17b]. This is a direct way to enforce retention and can be viewed as an optimization-based alternative to parameter importance. It is a canonical continual-learning method, but it is less established as a default for large-scale deep RL because repeated gradient constraints can be expensive and RL targets are themselves nonstationary.

Practical method selection

Situation	Recommended starting point	Why
Unknown task boundaries and compatible tasks	CLEAR-style replay	Strong, simple, and task-agnostic [Rol18]
Known tasks with limited raw-data storage	EWC or online importance regularization	Protects important parameters without retaining all transitions [Kir16]

Situation	Recommended starting point	Why
Continuously changing environments	Multi-timescale replay or consolidation	Avoids relying on sharp task boundaries [Kap18, Kap19]
Conflicting rewards for the same observation	Separate heads or modular policies	Represents incompatible objectives explicitly [Kes21]
Very strong retention requirements and few tasks	Progressive networks	Old policies are frozen and cannot be overwritten [Rus16]
Raw experience cannot be stored	Generative replay or world-model rehearsal	Recreates old training signals from a compact model [Atk18, Ket19b]

A sensible experimental ladder is therefore: sequential single-network training, standard replay, CLEAR-style replay, EWC, and a modular or multi-head model. If the task sequence is task-agnostic, the comparison should include a task-free replay baseline. If objectives conflict, a single-head replay baseline is not a sufficient control; the experiment should include a task-conditioned or multi-head upper bound.

Open limitations

The literature does not support a universal claim that one anti-forgetting method solves continual deep RL. Three gaps recur:

- **Evaluation mismatch:** many studies use clean task sequences with known boundaries, while real agents experience gradual or ambiguous change.
- **Transfer versus retention:** isolating parameters prevents forgetting but can reduce positive transfer; replay and shared representations can improve transfer but increase interference.
- **Benchmark mismatch:** sequential Atari tasks, changing-control benchmarks, and task-conflict environments measure different problems.

The most defensible conclusion is that catastrophic forgetting is best treated as a design problem involving memory, representation sharing, task inference, and objective compatibility. Replay is the canonical default, consolidation is a useful fixed-capacity complement, and modularity is necessary when the tasks are fundamentally incompatible.

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